

LAB 03

Ex 1.3:

```
ex1.3.cpp
1  #include<iostream>
2  #include<string>
3  using namespace std;
4  class Car{
5      public:
6          string brand;
7          string model;
8
9          void display() {
10             cout << "Brand: " << brand << endl;
11             cout << "Model: " << model << endl;
12         }
13 };
14
15 int main(){
16     Car c;
17     c.brand="toyota";
18     c.model="corolla";
19     c.display();
20     return 0;
21 }
```

```
cd "/home/rumaisa-abbasi/Desktop/LAB 03/" && g++ ex1.3.cpp -o
sktop/LAB 03/"ex1.3
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LAB 03$
AB 03/" && g++ ex1.3.cpp -o ex1.3 && "/home/rumaisa-abbasi/De
Brand: toyota
Model: corolla
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LAB 03$
```

LAB TASK 1.4:

```
1.4Labtask.cpp
1  #include <iostream>
2  #include<string>
3  using namespace std;
4
5  class Teacher {
6  public:
7      string name;
8      string subject;
9
10     void display() {
11         cout << "Name: " << name << endl;
12         cout << "Subject: " << subject << endl;
13     }
14 };
15
16 int main() {
17     Teacher t1, t2;
18
19     t1.name = "Ali";
20     t1.subject = "Mathematics";
21
22     t2.name = "Sara";
23     t2.subject = "Physics";
24
25     t1.display();
26     t2.display();
27
28     return 0;
```

```
Labtask
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LAB 03$ c
pp -o 1.4Labtask && "/home/rumaisa-abbasi/Desktop/LAB 03/"1.4L
Name: Ali
Subject: Mathematics
Name: Sara
Subject: Physics
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LAB 03$
```

Lab task 2.4:

```
2.3labtask.cpp
1  #include <iostream>
2  using namespace std;
3  class BankAccount {
4  private:
5      int balance;
6      int accountNumber; /*Private members cannot be accessed directly from outside the class.
7                          This protects sensitive data from accidental or unauthorized changes.*/
8  public:
9      void setBalance(int b) {
10         balance=b;
11     }
12     void setAccountNumber(int accNo) {
13         accountNumber=accNo;
14     }
15     void showBalance() {
16         cout <<"Balance: "<<balance<< endl;
17     }
18     void showAccountNumber() {
19         cout <<"Account Number: "<<accountNumber<< endl;
20     }
21 };
22 int main() {
23     BankAccount acc;
24
25     acc.setAccountNumber(12345);
26     acc.setBalance(10000);
27     acc.showAccountNumber();
28     acc.showBalance();
29     return 0;
30 }
```

```
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LAB 03$ cd "/t
/rumaisa-abbasi/Desktop/LAB 03/"2.3labtask
Account Number: 12345
Balance: 10000
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LAB 03$
```

Lab Task 3.3 :

```
#include <iostream>
using namespace std;

class Student
{
private:
    float cgpa;
public:
    void setCGPA(float c)
    {
        if (c >= 0.0 && c <= 4.0)
        {
            cgpa = c;
        }
        else
        {
            cout << "Invalid CGPA! CGPA must be between 0.0 and 4.0" << endl;
        }
    }
};

int main()
{
    Student s;
    s.setCGPA(3.5);
    s.setCGPA(4.8);
    return 0;
}
```

```
cd "/home/rumaisa-abbasi/Desktop/LAB 03/" && g++ 3.3.cpp -o 3.3 &&
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LAB 03$ cd "/"
.3 && "/home/rumaisa-abbasi/Desktop/LAB 03/"3.3
Invalid CGPA! CGPA must be between 0.0 and 4.0
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/LAB 03$
```

Lab Task 4.4:

.cpp file

.h file

```
#include <iostream>
#include "book.h"
using namespace std;

void Book::setTitle(string t) {
    title=t;
}
void Book::setAuthor(string a) {
    author=a;
}
void Book::setPages(int p) {
    if (p>0)pages=p;
    else pages=0;
}
string Book::getTitle() {
    return title;
}
string Book::getAuthor() {
    return author;
}
int Book::getPages() {
    return pages;
}
int main() {
    Book b;
    b.setTitle("One of us is dead");
    b.setAuthor("Jeneva Rose");
    b.setPages(464);
    cout << b.getTitle() << endl;
    cout << b.getAuthor() << endl;
    cout << b.getPages() << endl;
    return 0;
}
```

```
#ifndef BOOK_H
#define BOOK_H

#include <string>
using namespace std;

class Book
{
private:
    string title;
    string author;
    int pages;

public:
    void setTitle(string t);
    void setAuthor(string a);
    void setPages(int p);

    string getTitle();
    string getAuthor();
    int getPages();
};

#endif
```

```
● rumaisa-abbasi@abdullah-muaz-ThinkPad:
  asi/Desktop/LAB 03/"book
  One of us is dead
  Jeneva Rose
  464
○ rumaisa-abbasi@abdullah-muaz-ThinkPad:
```